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The deficits of isometric knee flexor strength in lengthened hamstring position after hamstring strain injury

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ABSTRACT

Objective: To investigate the effects of knee flexion angle on peak torque, rate of torque development (RTD) during isometric contraction and hamstring flexibility after hamstring strain injury (HSI).
Design: Cross-sectional.
Setting: Controlled laboratory research.
Participants: Fourteen male athletes with a history of HSI and 14 athletes without HSI (controls).
Main outcome measures: Hamstring flexibility was evaluated using active knee extension test. Isometric knee flexion peak torque and RTD were determined at 30°, 60°, and 90° of knee flexion measured by an isokinetic dynamometer.
Results: Individuals with a history of HSI had statistically significant, moderate deficits in isometric peak torque at 30° of knee flexion ($P = 0.037$; effect size = 0.55) in the HSI limb than in the uninjured limb, but not at 60° and 90° of knee flexion. In the control group, no significant differences in isometric peak torque at any angle were found between limbs. No differences in peak RTD and flexibility were found between limbs in both groups.
Conclusions: Isometric peak torque at 30° of knee flexion was lower in the injured limb than in the uninjured limb. Isometric strength deficits after HSI tended to be affected by lengthened hamstring angles.

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1. Introduction

Hamstring strain injury (HSI) is one of the most prevalent injuries in sports. HSI occurs in sports involving high-speed running, such as soccer (Ekstrand, Waldén & Hägglund, 2016; Ekstrand et al., 2020), Australian football (Orchard, Seward, & Orchard, 2013), and track and field (Edouard, Branco, & Alonso, 2016). Moreover, recurrent rates of HSI range from 13% to 63% (Ekstrand et al., 2016; Orchard et al., 2013; Verrall, Slavotinek, Barnes, Fon, & Spriggins, 2001), and reinjury leads to longer absence from sports compared with index injuries (Ekstrand et al., 2020). Previous history of HSI and older age are high-risk factors for HSI (Green, Bourne, Van Dyk, & Pizzari, 2020). Therefore, it is important to examine functional recovery after HSI to prevent reinjury.

The significance of isometric strength assessment for HSI and residual decrease after HSI has widely been indicated (De Vos et al., 2014; Timmins et al., 2015; Reurink et al., 2015; Maniar, Shield, Williams, Timmins, & Opar., 2016; Charlton et al., 2018; Hickey et al., 2018; Correia et al., 2020; Green et al., 2020). A greater decrease in isometric strength after HSI may contribute to reinjury (De Vos et al., 2014) and take longer time to return to sports (Maniar et al., 2016). Furthermore, the decrease in isometric peak knee flexion following HSI was reported to persist despite returning to full training and competition (Charlton et al., 2018; Hickey et al., 2018), contrary to other studies (Timmins et al., 2015; Reurink et al., 2015; Correia et al., 2020). Based on such conflicting evidence, the isometric strength assessment is not implemented as one of the criteria to return to sports following HSI (van der Horst, Backx, Goedhart, & Huisstede, 2017). Additionally, there are few

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studies on whether isometric muscle strength is associated with reinjury or not. Isometric testing is easy to implement in a clinical practice (Charlton et al., 2018). Further insight into isometric muscle strength after HSI may be important to clarify the high reinjury rate of HSI and introduce isometric testing as one of return to sports criteria after HSI in a clinical practice. During sprinting, the hamstring muscles are required to intensely produce large torque from the late swing phase to the initial contact phase (Sun, Wei, Zhong, Fu, Li, & Liu, 2015). Furthermore, the angular velocity of the knee extension was reported to reach approximately 1,000°/s (Kivi, Maraj, & Gervais, 2002). This indicates that hamstring muscles need to contract quickly and powerfully. The ability to produce explosive strength is often characterized as the rate of torque development (RTD), which has been analysed with torque-time curve at different time intervals (Maffiuletti et al., 2016). RTD was reported lower on the hamstring injured limb compared with the uninjured limb with anticipated eccentric contraction (Opar, Williams, Timmins, Dear, & Shield, 2013) and isometric contraction (Hickey et al., 2018), whereas no differences during isometric contraction were found in the injured limb compared with the control group (Correia., et al., 2020). It remains unclear whether the RTD after HSI was decreased compared with uninjured limb and no history of HSI or not. Therefore, further evidence regarding RTD after HSI should be investigated.

Some studies have reported that the knee flexors torque-angle curve was shifted toward more flexed, and the knee flexors torque decreased at the extended knee position with concentric and eccentric contraction after HSI (Brockett, Morgan, & Proske, 2004; Sole et al., 2011). Furthermore, it was speculated that a shorter optimum length of hamstring muscles might increase the risk of muscle damage (Brockett et al., 2004). These may indicate that the knee flexor strength after HSI should be assessed by several knee angles, whereas the deficits of muscle contractile function after HSI at only one angle might not detect the risk of reinjury. However, it remains unclear whether the isometric torque and RTD-angle relationship change after HSI because previous studies have examined isometric strength at one angle only (Timmins, Shield, Williams, Lorenzen, & Opar, 2015; Reurink et al., 2015; Correia et al., 2020). The between-limb differences of isometric knee flexor strength at muscle lengthened position could predict reinjury risk of HSI, but not at muscle shortened (De Vos et al., 2014). Therefore, the isometric peak strength and RTD should be assessed at different angles of knee flexion, which were lengthened muscles at the later stages of rehabilitation and during return to sports after HSI. With a recent meta-analysis, the deficits of isometric muscle strength at hamstring lengthened position were recovered on 20–50 days after injury, and there were no differences in isometric strength after returning to sport after HSI even at the extended knee position (Maniar et al., 2016). However, isometric muscle strength was measured at 0° of hip flexion. If the measurement was performed at hip neutral position, the deficits at muscle lengthened position might not be detected after HSI because hamstring muscles are two-joint muscles. Therefore, we thought that it is important to measure isometric muscle strength with the hip joint flexed. Considering the previous history of HSI has been consistently associated with a risk factor for future HSI (Green et al., 2020), the knowledge on the residual deficits after HSI might contribute to a better understanding regarding the reasons for high recurrent rates or an effective rehabilitation program.

Therefore, this study aimed to examine isometric knee flexor torque and rate of torque development at different knee flexor angles in participants with a history of HSI. We hypothesised that isometric knee flexor torque of the HSI limb would be lower close to the lengthened hamstring position compared to that of the uninjured limb. To the best of our knowledge, this is the first study to examine the effects of HSI on the isometric peak torque at different knee angles.

2. Methods

2.1. Participants

Fourteen young male athletes with a history of HSI and 14 young athletes without HSI were recruited (age range: 19–25 years). The sample size was estimated based on our pilot study because there was no investigation that explored the isometric strength at different knee angles in people with a history of HSI. G*Power software ver. 3.1.9.6 (University of Kiel, Kiel, Germany) was utilized to calculate the sample size needed for this study, in which a minimum sample size of 12 per group was required to reach a statistical power level ($1 - \beta$) of 0.80 based on a significant level of 0.05, a predicted effect size (η^2_p) of 0.15 for limb-by-angle interaction based on the results of our pilot study. Each group was entirely composed of college athletes, all of whom were regularly involved in the following sports: soccer, rugby football, baseball, track and field, and lacrosse. At the familiarization trial, we collected the participants' injury information, height, and body weight. HSI was defined as acute pain in the posterior thigh that was sustained during a match, training, or competition, that caused pain with contraction and/or stretching of the muscles (Charlton et al., 2018; Sole et al., 2011). The injury caused the participants to miss at least one match or required at least seven days of abstinence from normal activity. HSI was self-reported, and most cases were not diagnosed by physicians. All participants in the HSI group had fully returned to training and competitions. Participants with a history of injury that prevented them from playing sports at present, a history of knee or lumbopelvic surgery, and underlying neurological or cardiorespiratory disorders were excluded from the present study. The mean time from HSI was 22 months (range: 1 month to 7 years). Twelve participants in the HSI group and 11 in the control group were dominant on the right side. In the HSI group, 10 participants had sustained injuries to their dominant limb while four had sustained injuries to their non-dominant limb. Ten participants have sustained HSIs several times: Twice in 4 participants, more than 3 times in 6 participants.

All procedures were performed in accordance with the ethical standards of the institutional research committee, and the guidelines of 1964 Helsinki Declaration and its later amendments. The ethical committee of a local institutional review board approved the study (20-35-1). Written, informed consent was obtained from all individual participants included in the study.

2.2. Procedures

The participants were asked to come to a laboratory on two occasions, and each measurement was performed more than 48 h apart. The first visit was a familiarization session to practice the isometric testing procedure, measure hamstring flexibility, and collect anthropometric data. At the second visit, measurements were preceded by a warm-up session using a cycle ergometer for 5 min at 80 repetitions/min, five submaximal contractions, and two maximal contractions at concentric knee flexion at 60°/s from 0° (full extension) to 90° of knee flexion (Correia et al., 2020). The intensity of the submaximal contraction was determined by the participants. All participants performed isometric knee flexor strength measurements at knee flexion angles of 30°, 60°, and 90° in a seated position (85° of hip flexion). The order of the tested limb and knee flexion angles were randomised.

2.3. Flexibility measurement

Hamstring flexibility was determined using an active knee extension test (Gabbe, Bennell, Waiswelner, 2004; Sole et al., 2011). The right limb was measured first, followed by the left limb. The starting position was 90° of hip flexion. The participants were asked to extend one knee while maintaining the hip angle at 90° until the point of discomfort in the hamstring. An inclinometer (DL-155V, STS Inc., Japan) was placed in front of the tibia 15 cm below the tibial tuberosity, and the angle between the horizontal line and the tibia was calculated (Gabbe et al., 2004).

2.4. Isometric strength assessment

The maximal isometric knee flexor torque and rate of torque development (RTD) were determined using a Biodex System 3 isokinetic dynamometer (Biodex Medical System, Inc., Shirley, USA). The participants were assessed at 30°, 60°, and 90° of knee flexion in a seated position (85° of hip flexion). The hip joint was set at 85° of hip flexion, close to that of previous studies, which was reported at 90° of hip flexion (Correia et al., 2020; Sole et al., 2011; Brockett et al., 2004). This hip angle was determined from the angles between the seat and the back support as described in Biodex. The axis of rotation of the knee joint was aligned with the axis of rotation of the dynamometer, and the angles at which the shank segment was vertical was set to 90°. The chest, hip, and distal thigh of the test limb were firmly strapped, and we strapped these locations on every measurement. Three-second maximal voluntary isometric contraction (MVIC) trials were performed three times. A rest of 60 s was given between contractions for each trial, and the interval between two angles was approximately 2 min. The participants were instructed to contract as hard and fast as possible to obtain both isometric peak torque and peak RTD, respectively. The torque signal was sampled at 1,000 Hz and smoothed using a 12-Hz Butterworth fourth-order zero-lag filter; the effect of gravity was also adjusted (Correia et al., 2020; Opar et al., 2013). Isometric peak torque (Nm/kg) was calculated as the highest value from the filtered torque data during each MVIC, which was normalised by the body mass. Peak RTD (Nm/ms) was defined as the maximal slope at a time window of 200 ms from the onset to peak torque (Hicley et al., 2019) - a method reported to be the most reliable (Maffiuletti et al., 2016). The onset of muscle contraction was defined as the point at which the torque level reached 4 Nm and continued to increase past this threshold (Hickey et al., 2018; Opar et al., 2013). The highest isometric peak torque and peak RTD values were used for subsequent statistical analysis.

2.5. Statistical analysis

SPSS version 26.0 software (IBM Corp., Armonk, N.Y., USA) was used for all statistical analyses. Descriptive statistics were computed for all variables. Demographic data were compared between the HSI and control groups using independent t-tests. Dependent t-tests were performed for within-group comparisons of the active knee extension angle. A two-way repeated measures analysis of variances (limb-by-angle) was used to determine significant differences within each group for isometric peak torque and peak RTD. Post-hoc analysis using Bonferroni correction was employed to compare the differences between the limbs within each group. The significance level was set at $P < 0.05$. All data are presented as the mean ± SD. Cohen's d effect size for the *t*-test was used to assess the magnitude of differences within groups and between groups (Cohen's d). The effect size was classified as small ($d = 0.20–0.49$), moderate ($d = 0.50–0.79$), and large ($d \geqq 0.80$) (Batterham & Hopkins, 2006). Partial eta-squared values (η^2_p) were also reported to estimate the effect size for repeated measures.

3. Results

Table 1 shows the demographic data of both groups. The body mass, height, and body mass index were not different between the groups; age was significantly different.

As for the active knee extension test, no significant difference was found in the injured limb compared with the uninjured limb. In the control group, no differences were found in the active knee extension test (Table 2).

As for isometric peak torque (Table 2), a limb-by-angle interaction ($P = 0.020$; $\eta^2_p = 0.26$) and main effect for angle ($P < 0.010$; $\eta^2_p = 0.67$) were found among participants in the injured group, but no significant main effect was found for limb. Post-hoc analysis showed that the isometric peak torque of the injured limb was significantly lower than that of the uninjured limb at 30° of knee flexion ($P = 0.037$; $d = 0.55$), as shown in Fig. 1. At 60° and 90° knee flexion, no significant differences were observed between the injured and uninjured limbs. In the control group, there were main effects for angle ($P < 0.010$; $\eta^2_p = 0.68$), but no significant interaction and main effects for limb were found.

The peak RTD showed statistically significant main effects for angle in both groups (HSI group: $P < 0.010$; $\eta^2_p = 0.55$; control group: $P < 0.010$; $\eta^2_p = 0.51$) (Table 2). However, there was no significant interaction between limb-by-angle and main effects for limb in both groups.

4. Discussion

The present study investigated isometric peak torque and peak RTD at different knee flexion angles in individuals with and without a history of HSI. The main findings of this study were that isometric peak torque at 30° of knee flexion (at the lengthened hamstring angle) was lower in the previously injured limb than in the uninjured limb, but not at 60° and 90°. No significant difference in RTD was found at any knee flexion angle in the HSI group. This strength deficit persisted despite the return to full training practice and/or competition. This finding suggests that the deficits of isometric peak torque in participants with a history of HSI are more likely to be detected at the lengthened-hamstring angle.

In a recent meta-analysis, the deficits of isometric muscle strength at hamstring lengthened position were shown to recover within 20–50 days after injury, and there were no differences in isometric strength after the return to sport following HSI even at the extended knee position (Maniar et al., 2016). In contrast to previous studies, the present results showed that the isometric peak torque at 30° of knee flexion is significantly lower in the HSI limb than the contralateral limb, but not at 60° and 90°. This discrepancy may be explained by the angle of the hip joint when measuring the isometric peak torque. Previous studies measured the isometric knee flexor strength at 0° of knee flexion (Timmins et al., 2015), and at 15° of knee flexion (Reurink et al., 2015) with

Table 1
Demographic data (means ± standard deviations).

	Group		
	HSI	Control	P-values
Age (years)	22.0 ± 1.7	20.8 ± 1.1	0.041
Height (cm)	172.0 ± 5.6	172.3 ± 5.7	0.891
Body Mass (kg)	69.7 ± 11.9	67.7 ± 10.6	0.666
BMI (kg/m^2)	23.6 ± 4.2	22.7 ± 2.8	0.558
Time from injury (months)	21.7 ± 22.9		

HSI: hamstring strain injury; BMI: body mass index.

Table 2
Variables by limb and angle each group. (mean ± standard deviation).

Variables	HSI				Control			
	Injured	Uninjured	P-values	Cohen's d	Dominant	Non-dominant	P-values	Cohen's d
Active knee extension (deg)	47.7 ± 12.3	49.1 ± 11.1	0.351	0.121	46.7 ± 10.1	46.6 ± 8.0	0.978	0.011
Peak RTD (Nm/ms)								
30 deg	462.4 ± 100.5	503.9 ± 105.6	0.098	0.403	475.1 ± 140.3	461.8 ± 103.5	0.489	0.110
60 deg	450.3 ± 82.2	458.0 ± 81.3	0.721	0.093	442.9 ± 107.1	446.3 ± 79.5	0.846	0.040
90 deg	377.5 ± 56.9	391.9 ± 54.9	0.357	0.258	381.6 ± 111.0	387.6 ± 86.8	0.619	0.060

HSI: hamstring strain injury; RTD: rate of torque development.

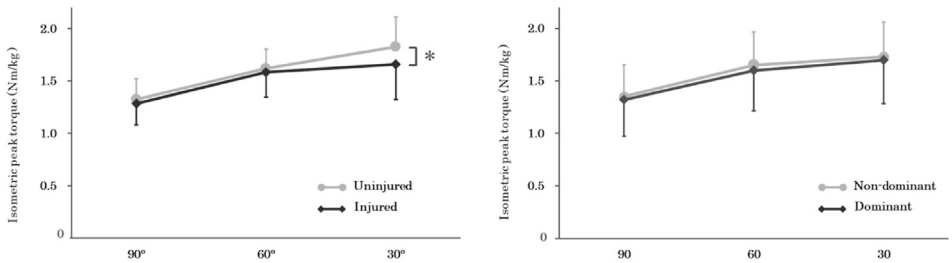

Fig. 1. Between-limb comparison of isometric peak torque at each knee flexion angle in the injured group (left) and the control group (right). Data illustrate mean and standard deviations. * denotes a significance differences between injured limb and uninjured limb.

0° of hip flexion. The length of hamstring muscles in previous studies (Timmins et al., 2015; Reurink et al., 2015) might be shorter than the present study position, and the difference in isometric peak torque was detected in our study (Kellis, 2018). Considering these observations, isometric peak torque measurement should be measured with the hip joint flexed to see the effects after HSI. Thus, the deficits of muscle function with isometric contraction are likely to be detected only at the lengthened muscle range, and the assessment should be encouraged in a clinical setting. Additionally, the deficits of isometric strength were one of the most important risk factors for reinjury (De Vos et al., 2014), and there should be further prospective studies to confirm whether the decreased isometric peak torque at a lengthened range of muscles is a risk factor for hamstring strain reinjury. Isometric strength training can be started from an early stage of HSI rehabilitation due to the low risk of inducing pain (Charlton et al., 2018). Furthermore, pain-threshold rehabilitation resulted in a great recovery of isometric strength compared with conventional pain-free rehabilitation (Hickey et al., 2020). Therefore, clinicians may be encouraged to instruct isometric training at the hamstring lengthened range even at an early stage of rehabilitation while being careful about pain to prevent persistent weakness of the muscle strength.

There were no significant differences in peak RTD of the participants with or without a history of HSI in this study. No differences in RTD with isometric contraction at 30° of knee flexion in a seated position between HSI limb and control group (Correia et al., 2020) were observed. Meanwhile, several previous studies (Hickey et al., 2018; Opar et al., 2013) have shown the RTD deficits on the hamstring injured limb compared with the uninjured limb. Hickey et al. (Hickey et al., 2018) found that large deficits were observed in a previously injured limb with RTD at 45° of knee and hip flexion using the original dynamometer. Moreover, Opar et al. (Opar et al., 2013) reported the reduced RTD and myoelectrical activity in injured muscles with eccentric contraction. Methodological differences (i.e., measurement position and contraction type) might have influenced the results between our findings and those from previous studies (Hickey et al., 2018; Opar et al., 2013). Furthermore, this study suggested that isometric peak torque rather than peak RTD is more likely to detect the decrease of muscle function after HSI, as indicated by Hickey et al. (Hickey et al., 2018).

This study demonstrated that the individuals with a history of HSI showed a decrease in isometric peak torque at the lengthened muscle range of the injured limb, although no side-differences in hamstring flexibility tests between the limbs were observed. A previous study showed that the hamstring flexibility and isometric strength were decreased immediately following HSI (Maniar et al., 2016). These findings indicated that the individuals with unilateral HSI, who had a decrease in isometric peak torque at the lengthened hamstring muscle range, might not be explained only by clinical measures of hamstring flexibility. Furthermore, this suggests that we should focus on other factors, such as improving the isometric peak torque at the lengthened muscle range. Muscle activity of biceps femoris long head during the late swing phase of sprinting was decreased in the injured limb than in the uninjured limb (Higashiahara et al., 2019; Daly, Persson, Twycross-Lewis, Woledge, & Morrissey., 2016). These findings suggested that the chronic functional deficits of the injured limb and muscle lengthened the angle, which supported our study results. The injured limb demonstrated less improvement of strength and fascicle length after the intervention of preseason exercise and across the Australian football league season (Opar, Williams, Timmins, Hickey, Duhig, & Shield, 2015; Timmins et al., 2017), which may cause functional deficits of lengthened muscles after HSI. The deficits of function at the lengthened hamstring muscles may be due to structural (i.e., fascicle length) (Timmins et al., 2015) and neurological factors (i.e., EMG or neuromuscular inhibition) (Fyfe et al., 2013; Sole et al., 2011). Further investigations are warranted to explore the mechanisms of decreased isometric peak torque at the lengthened hamstring muscles using EMG and ultrasonography to strengthen our findings.

There were some limitations to this study. First, the injury was self-reported, and most participants were not diagnosed by a physician; therefore, the location and severity of the injury were not specified. Furthermore, most participants (64%) did not undergo rehabilitation, which may affect the results. Moreover, all participants in this study were males. If the participant were females, the results may be different. Second, there is a wide variation from the time of HSI (1 month–7 years), and caution should be

taken in interpreting the results. Third, hip extensors might contribute to the torque production of knee flexion, but the sEMG was not determined in this study. To minimize the compensatory movement, each strap was fastened on every measurement. Fourth, this was a cross-sectional study, so we could not clarify whether these findings were the result of or a cause of HSI. Fifth, we did not determine architectural change and muscle activity; therefore, we could not explore the mechanism of the isometric muscle strength deficits in the outer range. Finally, we did not control the dominant side of participants in this study. Four participants' (29%) injured side were not the dominant side. Furthermore, we could not match the play of sports among the participants. If these participants were excluded from the analysis, the numbers of participants expected from a priori power analysis would have been lacking. Thus, we could not compare the injured and control groups. In the future, it is necessary to adjust for these confounding factors, and the comparison between the injured and control groups should be investigated.

5. Conclusion

Our findings showed that the deficits in isometric peak muscle strength of the HSI limb were found at the lengthened muscle position, but not in the unstretched position. No differences in RTD were observed between the injured and uninjured limbs. These results indicate that clinicians should assess and strengthen the isometric peak torque of knee flexors at the hamstring lengthened position after HSI.

Ethical approval

This study was approved by the local institutional review board (approval number: 20-35-1).

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Declaration of competing interest

None declared.

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